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Fiber optic connector

Abstract

A connector includes a ferrule assembly having a ferrule, a hub and a spring, the ferrule having a distal face accessible at a distal end of the connector housing, the ferrule being movable in a proximal direction relative to the connector housing. The distal and proximal positions are separated by an axial displacement distance. The ferrule proximal movement is against the spring's bias. The cable of the assembly includes an optical fiber contained within a jacket and also a strength layer between the fiber and the jacket that is anchored to the connector housing. The fiber extends through a fiber from the proximal end of the connector housing to the ferrule. The fiber has a distal portion potted within the ferrule. The fiber passage has a fiber take-up region configured to take-up an excess length of the fiber corresponding to the ferrule axial displacement.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/110,854, filed Dec. 3, 2020, now U.S. Pat. No. 11,782,224; which is a continuation of U.S. patent application Ser. No. 16/696,629, filed Nov. 26, 2019, now U.S. Pat. No. 10,859,771; which is a continuation of U.S. patent application Ser. No. 16/204,672, filed Nov. 29, 2018, now U.S. Pat. No. 10,495,822; which is a continuation of U.S. patent application Ser. No. 15/837,290, filed Dec. 11, 2017, now U.S. Pat. No. 10,146,011; which is a continuation of U.S. patent application Ser. No. 15/357,030, filed Nov. 21, 2016, now U.S. Pat. No. 9,841,566; which is a continuation of U.S. patent application Ser. No. 14/858,900, filed Sep. 18, 2015, now U.S. Pat. No. 9,500,813; which is a continuation of U.S. patent application Ser. No. 14/154,352, filed Jan. 14, 2014, now U.S. Pat. No. 9,151,904; which is a continuation of U.S. patent application Ser. No. 13/420,286, filed Mar. 14, 2012, now U.S. Pat. No. 8,636,425, which claims the benefit of U.S. Provisional Patent Application Serial Nos. 61/510,711, filed Jul. 22, 2011; and 61/452,953, filed Mar. 15, 2011, which applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure relates generally to optical fiber communication systems. More particularly, the present disclosure relates to fiber optic connectors used in optical fiber communication systems.

BACKGROUND

(2) Fiber optic communication systems are becoming prevalent in part because service providers want to deliver high bandwidth communication capabilities (e.g., data and voice) to customers. Fiber optic communication systems employ a network of fiber optic cables to transmit large volumes of data and voice signals over relatively long distances. Optical fiber connectors are an important part of most fiber optic communication systems. Fiber optic connectors allow two optical fibers to be quickly optically connected without requiring a splice. Fiber optic connectors can be used to optically interconnect two lengths of optical fiber. Fiber optic connectors can also be used to interconnect lengths of optical fiber to passive and active equipment.

(3) A typical fiber optic connector includes a ferrule assembly supported at a distal end of a connector housing. A spring is used to bias the ferrule assembly in a distal direction relative to the

connector housing. The ferrule functions to support an end portion of at least one optical fiber (in the case of a multi-fiber ferrule, the ends of multiple fibers are supported). The ferrule has a distal end face at which a polished end of the optical fiber is located. When two fiber optic connectors are interconnected, the distal end faces of the ferrules abut one another and the ferrules are forced proximally relative to their respective connector housings against the bias of their respective springs. With the fiber optic connectors connected, their respective optical fibers are coaxially aligned such that the end faces of the optical fibers directly oppose one another. In this way, an optical signal can be transmitted from optical fiber to optical fiber through the aligned end faces of the optical fibers. For many fiber optic connector styles, alignment between two fiber optic connectors is provided through the use of an intermediate fiber optic adapter.

(4) A fiber optic connector is often secured to the end of a corresponding fiber optic cable by anchoring strength members of the cable to the connector housing of the connector. Anchoring is typically accomplished through the use of conventional techniques such as crimps or adhesive. Anchoring the strength members of the cable to the connector housing is advantageous because it allows tensile load applied to the cable to be transferred from the strength members of the cable directly to the connector housing. In this way, the tensile load is not transferred to the ferrule assembly of the fiber optic connector. If the tensile load were to be applied to the ferrule assembly, such tensile load could cause the ferrule assembly to be pulled in a proximal direction against the bias of the connector spring thereby possibly causing an optical disconnection between the connector and its corresponding mated connector. Fiber optic connectors of the type described above can be referred to as pull-proof connectors.

(5) As indicated above, when two fiber optic connectors are interconnected together, the ferrules of the two connectors contact one another and are respectively forced in proximal directions relative to their housings against the bias of their respective connector springs. In the case of pull-proof connectors, such proximal movement of the ferrules causes the optical fibers secured to the ferrules to move proximally relative to the connector housings and relative to the jackets of the fiber optic cables secured to the connectors. To accommodate this relative proximal movement of the optical fibers, the fiber optic cables typically have sufficient interior space to allow the optical fibers to bend in a manner that does not compromise signal quality in a meaningful way. Typically, the bending comprises “macro-bending” in which the bends have radii of curvatures that are larger than the minimum bend radius requirements of the optical fiber.

(6) A number of factors are important with respect to the design of a fiber optic connector. One aspect relates to ease of manufacturing and assembly. Another aspect relates to connector size and the ability to provide enhanced connector/circuit densities. Still another aspect relates to the ability to provide high signal quality connections with minimal signal degradation.

SUMMARY

(7) One aspect of the present disclosure relates to a fiber optic connector having features that facilitate connector assembly. For example, such features can include structures for enhancing guiding optical fibers into a connector during assembly, and for facilitating applying epoxy into a ferrule of a connector during assembly.

(8) Another aspect of the present disclosure relates to fiber optic connectors having features that prevent unacceptable bending of an optical fiber when ferrules of the connectors are moved proximally relative to the connector housings as two connectors are coupled together. In certain embodiments, the connectors can include space for accommodating macro-bending of the optical fibers within the connector housings.

(9) A variety of additional aspects will be set forth in the description that follows. The aspects relate to individual features and to combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective, exploded view of a fiber optic connector in accordance with the principles of the present disclosure.
- (2) FIG. 2 is a cross-sectional view that longitudinally bisects the fiber optic connector of FIG. 1.
- (3) FIG. 3 is a perspective view of a rear housing of the fiber optic connector of FIG. 1.
- (4) FIG. 4 is a cross-sectional view that longitudinally bisects the rear housing of FIG. 3.
- (5) FIG. 5 is a perspective view showing a first end of a first insertion cap that can be used with the fiber optic connector of FIG. 1.
- (6) FIG. 6 is a perspective view showing a second end of the insertion cap of FIG. 5.
- (7) FIG. 7 is a cross-sectional view that longitudinally bisects the insertion cap of FIGS. 5 and 6.
- (8) FIG. 8 is a perspective view showing a first end of a second insertion cap that can be used with the fiber optic connector of FIG. 1.
- (9) FIG. 9 is a perspective view showing a second end of the insertion cap of FIG. 8.
- (10) FIG. 10 is a cross-sectional view that bisects the insertion cap of FIGS. 8 and 9.
- (11) FIG. 11 is a perspective view showing a first end of a strain relief boot of the fiber optic connector of FIG. 1.
- (12) FIG. 12 is a perspective view showing a second end of the strain relief boot of FIG. 11.
- (13) FIG. 13 is a cross-sectional view that longitudinally bisects the strain relief boot of FIGS. 11 and 12.
- (14) FIG. 14 is an exploded, perspective view of a second fiber optic connector in accordance with the principles of the present disclosure.
- (15) FIG. 15 is a cross-sectional view that longitudinally bisects the fiber optic connector of FIG. 14.
- (16) FIG. 16 is a perspective view showing a first side of a half-piece of a rear housing of the fiber optic connector of FIG. 14.
- (17) FIG. 17 is a perspective view showing a second side of the half-piece of FIG. 16.
- (18) FIG. 18 is side view showing the second side of the half-piece of FIGS. 16 and 17.
- (19) FIG. 19 is a perspective view showing a first end of a first insertion cap that can be used with the fiber optic connector of FIG. 14.
- (20) FIG. 20 is a perspective view showing a second end of the insertion cap of FIG. 19.
- (21) FIG. 21 is a cross-sectional view that longitudinally bisects the insertion cap of FIGS. 19 and 20.
- (22) FIG. 22 is a perspective view showing a first end of a second insertion cap that can be used with the fiber optic connection of FIG. 14.
- (23) FIG. 23 is a perspective view showing a second end of the insertion cap of FIG. 22.
- (24) FIG. 24 is a cross-sectional view that longitudinally bisects the insertion cap of FIGS. 22 and 23.
- (25) FIG. 25 is a cross-sectional view that longitudinally bisects a prior art fiber optic adapter.
- (26) FIG. 26 is a cross-sectional view taken along section line 26-26 of FIG. 2.
- (27) FIG. 27 is a top view of a prior art LC style fiber optic connector.
- (28) FIG. 28 is a cross-sectional view that longitudinally bisects the fiber optic connector of FIG. 27.
- (29) FIG. 29 is a perspective, exploded view of a third fiber optic connector having features with inventive aspects in accordance with the principles of the present disclosure.
- (30) FIG. 30 is a partially assembled perspective view of the fiber optic connector of FIG. 29.
- (31) FIG. 31 is a fully assembled perspective view of the fiber optic connector of FIG. 29.
- (32) FIG. 32 is a top view of the fiber optic connector of FIG. 29.

- (33) FIG. 33 is a cross-sectional view that longitudinally bisects the fiber optic connector of FIG. 29.
- (34) FIG. 34 illustrates a perspective view of two of the fiber optic connectors of FIG. 29 coupled to a duplex LC fiber optic adapter.
- (35) FIG. 35 is a side view of the fiber optic connectors coupled to a duplex LC fiber optic adapter of FIG. 34.
- (36) FIG. 36 is a top view of the fiber optic connectors coupled to a duplex LC fiber optic adapter of FIG. 34.
- (37) FIG. 37 illustrates a perspective view of two of the fiber optic connectors of FIG. 29 coupled together by a clip to form a duplex fiber optic connector.
- (38) FIG. 38 is a top view of the duplex fiber optic connector of FIG. 37.
- (39) FIG. 39 is a perspective view of a front housing of the fiber optic connector of FIG. 29.
- (40) FIG. 40 is a side view of the front housing of the fiber optic connector of FIG. 39, with a portion of the front housing broken-away to illustrate the internal configuration thereof.
- (41) FIG. 41 is a perspective view of a rear housing of the fiber optic connector of FIG. 29.
- (42) FIG. 42 is a cross-sectional view that longitudinally bisects the rear housing of FIG. 41.
- (43) FIG. 43 is a cross-sectional view that longitudinally bisects the insertion cap of the fiber optic connector shown in FIG. 29.
- (44) FIG. 44 is a perspective view of a strain relief boot of the fiber optic connector of FIG. 29.
- (45) FIG. 45 is a cross-sectional view that longitudinally bisects the strain relief boot of FIG. 41.
- (46) FIG. 46 is a perspective, exploded view of a fourth fiber optic connector having features with inventive aspects in accordance with the principles of the present disclosure.
- (47) FIG. 47 is a partially assembled perspective view of the fiber optic connector of FIG. 46.
- (48) FIG. 48 is a fully assembled perspective view of the fiber optic connector of FIG. 46.
- (49) FIG. 49 is a top view of the fiber optic connector of FIG. 46.
- (50) FIG. 50 is a cross-sectional view that longitudinally bisects the fiber optic connector of FIG. 46.
- (51) FIG. 51 is a perspective view of a rear housing of the fiber optic connector of FIG. 46.
- (52) FIG. 52 is a front view of the rear housing of FIG. 51.
- (53) FIG. 53 is a cross-sectional view taken along line 53-53 of FIG. 52.
- (54) FIG. 54 is a cross-sectional view taken along line 54-54 of FIG. 53.
- (55) FIG. 55 is a cross-sectional view taken along line 55-55 of FIG. 54.
- (56) FIG. 56 is a perspective view of an insertion cap that can be used with the fiber optic connector of FIG. 46.
- (57) FIG. 57 is cross-sectional view that bisects the insertion cap of FIG. 56.
- (58) FIG. 58 is a cross-sectional view taken along line 58-58 of FIG. 57.
- (59) FIG. 59 is a cross-sectional view taken along line 59-59 of FIG. 57.
- (60) FIG. 60 is a rear perspective view of an example embodiment of a crimp sleeve that might be used to anchor the optical fiber to the connector housing of a fiber optic connector.
- (61) FIG. 61 is a rear view of the crimp sleeve of FIG. 60.
- (62) FIG. 62 is a cross-sectional view taken along lines 62-62 of FIG. 61.
- (63) FIG. 63 is a rear perspective view of another example embodiment of a crimp sleeve that might be used to anchor the optical fiber to the connector housing of a fiber optic connector.
- (64) FIG. 64 is a rear view of the crimp sleeve of FIG. 63.
- (65) FIG. 65 is a cross-sectional view taken along lines 65-65 of FIG. 61.

DETAILED DESCRIPTION

(66) FIGS. 1 and 2 illustrate a first fiber optic connector 20 in accordance with the principles of the present disclosure. The fiber optic connector 20 has a total length L_i that extends from a distal end 22 of the fiber optic connector 20 to a proximal end 24 of the fiber optic connector 20. The fiber optic connector 20 includes a ferrule assembly 26 that mounts adjacent the distal end 22 of the fiber

optic connector **20**. The ferrule assembly includes a ferrule **28**, a hub **30** and a spring **31**. The ferrule assembly **26** mounts at least partially within a connector housing **32** including a distal housing portion **34** that interconnects with a proximal housing portion **36**. In one embodiment, the distal housing portion **34** snaps over ribs **37** provided on the proximal housing portion **36** to interlock the two housing portions together. The fiber optic connector **20** also includes a release sleeve **38** that slidably mounts over the connector housing **32**. The fiber optic connector **20** further includes an insertion cap **40A** that mounts inside a proximal end **42** of the proximal housing portion **36** and a crimp sleeve **44** that mounts around the exterior of the proximal end **42** of the proximal housing portion **36**. The proximal end **24** of the fiber optic connector **20** is configured to receive, anchor and provide strain relief/bend radius protection to a fiber optic cable **46**. The fiber optic cable **46** includes a jacket **48** surrounding at least one optical fiber **50**. The fiber optic cable **46** also includes a strength layer **52** formed by a plurality of strength members (e.g., reinforcing fibers such as aramid yarn/Kevlar) positioned between the optical fiber **50** and the jacket **48**. A distal end portion of the strength layer **52** is crimped between the crimp sleeve **44** and the exterior surface of the proximal end **42** of the proximal housing portion **36** so as to anchor the strength layer **52** to the connector housing **32**. The optical fiber **50** is routed through the total length L_i of the fiber optic connector **20** and includes a distal portion **54** secured within the ferrule **28**. The fiber optic connector **20** further includes a strain relief boot **56** mounted at the proximal end **24** of the fiber optic connector **20** for providing strain relief and bend radius protection to the optical fiber **50**.

(67) It will be appreciated that the fiber optic connector **20** is adapted to be mechanically coupled to a like fiber optic connector by an intermediate fiber optic adapter. FIG. 25 shows an example fiber optic adapter **58** that can be used to couple two of the fiber optic connectors **20** together. The fiber optic adapter **58** includes an adapter housing **59** defining opposite, coaxially aligned ports **60**, **62** for receiving two of the fiber optic connectors desired to be coupled together. The fiber optic adapter **58** also includes an alignment sleeve **64** for receiving and aligning the ferrules **28** of the fiber optic connectors desired to be connected together. The fiber optic adapter **58** further includes latches **66** for mechanically retaining the fiber optic connectors **20** within their respective ports **60**, **62**. The latches **66** can be configured to engage shoulders **68** provided on the distal housing portions **34** of the fiber optic connectors **20** being coupled together. Further details regarding the fiber optic adapter **58** can be found in U.S. Pat. No. 5,317,633, which is hereby incorporated by reference in its entirety.

(68) In the depicted embodiment of FIG. 1, the release sleeve **38** is shown as a conventional SC release sleeve. When the release sleeve **38** is mounted on the connector housing **32**, the release sleeve **38** is free to slide back-and-forth in distal and proximal directions relative to the connector housing **32** along a central longitudinal axis **70** of the fiber optic connector **20**. When the fiber optic connector **20** is inserted within one of the ports **60**, **62** of the fiber optic adapter **58**, the keying rail **72** provided on the release sleeve **38** ensures that the fiber optic connector **20** is oriented at the appropriate rotational orientation relative to the fiber optic adapter **58**. When the fiber optic connector **20** is fully inserted within its corresponding port **60**, **62**, the latches **66** snap into a latching position in which the latches engage the shoulders **68** of the connector housing **32** to prevent the fiber optic connector **20** from being proximally withdrawn from the port **60**, **62**. The release sleeve **38** is provided to allow the fiber optic connector **20** to be selectively withdrawn from its respective port **60**, **62**. Specifically, by pulling the release sleeve **38** in a proximal direction, ramps **74** of the release sleeve disengage the latches **66** of the fiber optic adapter **58** from the shoulders **68** of the fiber optic connector **20** thereby allowing the fiber optic connector **20** to be proximally withdrawn from its respective port **60**, **62**.

(69) Referring to FIG. 2, the ferrule **28** of the ferrule assembly **26** includes a distal end **76** and a proximal end **78**. The distal end **76** projects distally outwardly beyond a distal end of the connector housing **32** and the proximal end **78** is secured within the ferrule hub **30**. When the connector housing **32** is assembled as shown at FIG. 2, the ferrule hub **30** and the spring **31** are captured

between the distal housing portion **34** and the proximal housing portion **36** of the connector housing **32**. As so configured, the spring **31** is configured to bias the ferrule **28** in a distal direction relative to the connector housing **32**. When two of the fiber optic connectors **20** are interconnected, their ferrules **28** are forced to move in proximal directions relative to their respective connector housings **34** against the bias of their respective springs **31**. The movement is along the central axes **70** of the mated fiber optic connectors **20**.

(70) Referring to FIGS. **2** and **26**, the jacket **48** of the fiber optic cable **46** preferably has a relatively small outer diameter $D_{sub.1}$. In certain embodiments, the outer diameter $D_{sub.1}$ can be less than 2 millimeters, or less than 1.5 millimeters, less than equal to about 1.2 millimeters. In certain embodiments, the optical fiber **50** within the jacket **48** can include a core **90**, a cladding layer **92** surrounding the core and one or more coating layers **94** surrounding the cladding layer **92**. In certain embodiments, the core **90** can have an outer diameter of about 10 microns, the cladding layer **92** can have an outer diameter of about 125 microns, and the one or more coating layers **94** can have an outer diameter in the range of about 240 to 260 microns. The strength layer **52** provides tensile reinforcement to the cable **46**. The strength layer **52** relatively closely surrounds the coating layer **94** of the optical fiber **50**. In addition to providing tensile strength to the cable **46**, the strength layer **52** also functions as a separator for separating the optical fiber **50** from the outer jacket **48**. In certain embodiments, no buffer layer or buffer tube is provided between the coating layer **94** of the optical fiber **50** and the strength layer **52**. Further details regarding the fiber optic cable **46** can be found in U.S. Pat. No. 8,548,293, which is hereby incorporated by reference in its entirety.

(71) As shown at FIG. **2**, the optical fiber **50** extends through the total length L_i of the fiber optic connector **20**. For example, the optical fiber **50** extends through the strain relief boot **56**, the insertion cap **40A**, the connector housing **32** and the ferrule **28**. In certain embodiments, a portion of the optical fiber **50** extending proximally from the ferrule **28** through the fiber optic connector **20** to the jacketed portion of the fiber optic cable **46** includes only the core **90**, the cladding layer **92** and the one or more coating layers **94**. The portion of the optical fiber **50** extending through the ferrule **28** typically only includes the core **90** and the cladding layer **92**. A distal most end face of the optical fiber **50** is preferably polished as is conventionally known in the art.

(72) As shown at FIG. **2**, the insertion cap **40A** (see FIGS. **5-7**) is mounted within the proximal end **42** of the proximal housing portion **36** of the connector housing **32**. The insertion cap **40A** has an inner diameter $D_{sub.2}$ sized to correspond with the outer diameter of the coating layer **94**. In alternative embodiments, it may be desirable to cover/protect the portion of the optical fiber **50** extending through the connector housing **32** with a protective layer such as a 900 micron tube (e.g., a 900 micron furcation tube). To accommodate such a protective tube, the insertion cap **40A** can be replaced with an insertion cap **40B** (see FIGS. **8-10**) having an inner diameter $D_{sub.3}$ that is larger than the inner diameter $D_{sub.2}$. In certain embodiments, inner diameter $D_{sub.3}$ can correspond to the outer diameter of protective buffer tube provided about the coating layer **94** of the optical fiber **50** within the connector housing **32**.

(73) The fiber optic connector **20** is a pull-proof connector in which the strength layer **52** of the fiber optic cable **46** is anchored to the connector housing **32** thereby preventing tensile loads from being transferred to the ferrule assembly **26**. Because of this configuration, movement of the ferrule **28** in a proximal direction relative to the connector housing **32** causes the optical fiber **50** to be forced/displaced in a proximal direction relative to the connector housing **32** and the jacket **48** of the fiber optic cable **46**. In the depicted embodiment, the ferrule **28** has a maximum axial displacement AD in the proximal direction during the connection process. The axial displacement AD creates an excess fiber length having a length equal to the length of the axial displacement AD . In certain embodiments, the maximum axial displacement AD can be 0.035 inches.

(74) With regard to the axial displacement AD described above, it is significant that the relatively small diameter of the fiber optic cable **46** and the lack of open space within the interior of the

jacket **48** do not allow the cable **46** to readily accommodate acceptable macrobending of the optical fiber **50** within the jacket **48** when the ferrule **28** is forced in a proximal direction relative to the connector housing **32**. Therefore, to prevent signal degradation related to microbending caused by the axial displacement of the optical fiber **50** in the proximal direction, the connector **20** is itself preferably configured to take-up the excess fiber length corresponding to the axial displacement. To take-up the excess fiber length, the fiber optic connector **20** includes features that encourage a controlled, predictable and repeatable macrobend of the optical fiber **50** within the connector housing **32** when the ferrule **28** is forced in a proximal direction relative to the connector housing **32**. In this way, the fiber optic connector **20** itself accommodates the acceptable macrobending of the optical fiber **50** such that the optical fiber **50** does not need to slide within the jacket **48** of the fiber optic cable **46** and does not require the optical fiber **52** to macro or microbend within the jacket **48** of the fiber optic cable **46** when the ferrule **28** is forced in a proximal direction relative to the connector housing **32**.

(75) To prevent unacceptable signal degradation, the fiber optic connector **20** is preferably designed to take-up the optical fiber length corresponding to the axial displacement AD. For example, referring to FIG. 2, the connector housing **32** includes a fiber take-up region **100** that extends generally from a proximal end of the spring **31** to the proximal end **42** of the proximal housing portion **36**. The fiber take-up region **100** includes a passage **101** that extends along the axis **70**. As shown at FIG. 2, the passage **101** has an intermediate section **102**, a distal section **104** and a proximal section **106**. The intermediate section **102** has an enlarged transverse cross-sectional area as compared to the transverse cross-sectional areas of the distal and proximal sections **104**, **106**. The transverse cross-sectional areas are taken along planes perpendicular to the longitudinal axis **70** of the connector **20**. The distal section **104** and the intermediate section **102** are defined by the proximal housing portion **36** (see FIG. 4). The distal section **104** of the passage **101** has a necked configuration with a neck portion **104a** positioned between transition portions **104b** and **104c**. The neck portion **104a** defines a minimum cross-dimension CD1 (e.g., an outer diameter) and minimum transverse cross-sectional area of the distal section **104**. The transition portion **104b** provides a gradual reduction in transverse cross-sectional area (i.e., a funnel or taper toward the longitudinal axis **70**) as the transition portion **104b** extends from the intermediate section **102** of the passage **101** toward the neck portion **104a**. The transition portion **104c** provides a gradual increase in transverse cross-sectional area (i.e., a funnel or taper away from the longitudinal axis **70**) as the transition portion **104c** extends from the neck portion **104a** toward the spring **31**.

(76) The proximal section **106** of the passage **101** is defined by the inside of the insertion cap **40A** or the insertion cap **40B** (depending on which one is selected). For ease of explanation, the description herein will primarily refer to the insertion cap **40A** (see FIGS. 5-7). A minimum cross-dimension CD2 (e.g., an outer diameter) of the proximal section **106** is defined near a proximal end of the insertion cap **40A**. The proximal section **106** includes a transition **106a** that provides a reduction in transverse cross-sectional area as the transition **106a** extends in a proximal direction from the intermediate section **102** of the passage **101** toward the minimum cross-dimension CD2. A chamfer **109** at the proximal end of the insertion cap **40A** provides an increase in transverse cross-sectional area as the chamfer **109** extends proximally from the minimum cross-dimension CD2. The chamfer **109** can assist in providing bend radius protection with respect to the fiber passing through the insertion cap **40A**. It will be appreciated that by using the insertion cap **40B**, the minimum diameter provided by the insertion cap can be enlarged so as to accommodate a productive buffer tube covering the optical fiber **50** within the passage **101**.

(77) In certain embodiments, the minimum cross-dimension CD1 is greater than the minimum cross-dimension CD2. In other embodiments, the minimum cross-dimension CD1 is at least twice as large as the minimum cross-dimension CD2. In other embodiments, the minimum cross-dimension CD1 is generally equal to the minimum cross-dimension CD2. In still further embodiments, a maximum cross-dimension CD3 of the passage **101** is at least 1.5 times or 2 times

as large as the minimum cross-dimension CD1. In still other embodiments, the maximum cross-dimension CD3 of the passage **101** is at least 2, 3 or 4 times as large as the minimum cross-dimension CD2.

(78) It will be appreciated that the length and transverse cross-sectional dimensions of the fiber take-up region **100** are selected to accommodate the excess length of fiber corresponding to the axial displacement distance AD. When the ferrule **28** is pushed in a proximal direction, the configuration of the fiber take-up region **100** causes the optical fiber **50** to move from a generally straight path SP along the axis **70** to a path that follows generally along a single macrobend **120** (shown at FIG. 2) that extends along the surface of the fiber take-up region **100** from the distal section **104** through the intermediate section **102** to the proximal section **106**. The increase in length between the straight path and the curved path equals the axial displacement distance AD. The transitions **104b**, **106a** provided at the proximal and distal sections **104**, **106** of the passage **101** help to encourage the fiber to form the single microbend in a predictable, repeatable manner as the ferrule **28** is forced in a proximal direction relative to the connector housing **32** during a connection process. In certain embodiments, the fiber take-up region is configured to take up at least 0.015 inches, or at least 0.025 inches or at least 0.035 inches of excess fiber length.

(79) In addition to the advantages provided above, the transition **104b** also facilitates assembly of the fiber optic connector **20**. Specifically, during assembly, the optical fiber **50** is inserted in a distal direction through the proximal end **42** of the connector housing **32** and is directed through the length of the connector housing into the ferrule **28**. The transition **104b** assists in guiding the fiber **50** into the ferrule **28** during the fiber insertion process.

(80) Referring to FIG. 7, the insertion cap **40A** includes a sleeve portion **110** having a cylindrical outer surface that fits inside the proximal end **42** of the connector housing **32**. The insertion cap **40A** also includes a flange **112** at a proximal end of the sleeve portion **110**. The flange **112** projects radially outwardly from the cylindrical outer surface of the sleeve portion **110** and forms a proximal end of the insertion cap **40A**. The flange **112** abuts against the proximal end **42** of the connector housing **32** when the insertion cap **40A** is inserted therein. The inside of the insertion cap **40A** defines the proximal section **106** of the passage **101** which extends in a proximal to distal direction through the insertion cap **40A**. The insertion cap **40B** has a similar configuration as the insertion cap **40A**, except the minimum inner cross-dimension CD2 (e.g., inner diameter) of the insertion cap **40B** is larger than the minimum cross-dimension CD2 of the insertion cap **40A** so as to better accommodate a protective tube covering the coated fiber **50** within the connector housing **32**.

(81) The use of the insertion cap **40A** or the insertion cap **40B** allows the proximal end **42** of the connector housing **32** to have a relatively large open transverse cross-sectional area which corresponds to the maximum cross-dimension CD3 of the passage **101**. This large transverse cross-sectional area is advantageous because it facilitates delivering potting material (e.g., and adhesive material such as epoxy) to the back side of the ferrule **28** during assembly for potting the fiber **50** within the ferrule **28**. Typically, a needle can be used to deliver potting material to the ferrule **28**. The large cross-sectional area provides better access for allowing a needle to be inserted through the proximal end of the connector housing **32** to accurately injecting potting material into the ferrule **28**.

(82) Referring to FIG. 1, the crimp sleeve **44** of the fiber optic connector **20** includes a sleeve portion **140** and a stub portion **142** that projects proximally outwardly from a proximal end of the sleeve portion **140**. A radial in-step **141** is provided between the sleeve portion **140** and the stub portion **142** such that the sleeve portion **140** has a larger diameter than the stub portion **142**. A passage extends axially throughout the length of the crimp sleeve **44**. The passage has a smaller diameter through the stub portion **142** and a larger diameter through the sleeve portion **140**. When the fiber optic connector **20** is assembled, the sleeve portion **140** is crimped about the exterior surface of the connector housing **32** adjacent the proximal end **42** of the connector housing **32** (see

FIG. 2). The exterior surface of the connector housing **32** can be textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the crimp on the housing **32**. Preferably, a distal portion of the strength layer **52** of the fiber optic cable **46** is crimped between the sleeve portion **140** and the exterior surface of the connector housing **32** such that the strength layer **52** of the cable **46** is anchored relative to the connector housing **32**.

(83) In certain embodiments (e.g., as shown in FIG. 1), the sleeve portion **140** of the crimp sleeve may include an annular rib **143** on an exterior surface thereof. The annular rib **143** may provide additional material for the crimp sleeve **44** at spots or regions that will tend to deform when the crimp sleeve **44** is crimped at the sleeve portion **140**.

(84) The stub portion **142** fits within a pocket **144** provided within the strain relief boot **56**. The stub portion **142** coaxially aligns with the central longitudinal axis **70** of the fiber optic connector **20**. The insertion cap **40A** is captured between the proximal end **42** of the connector housing **32** and the crimp sleeve **44**. In this way, the crimp sleeve **44** assists in retaining the insertion cap **40A** in the proximal end **42** of the connector housing **32**. The insertion cap **40A** can also be held within the connector housing **22** by an adhesive material such as epoxy.

(85) In certain embodiments, it can be advantageous to crimp the stub portion **142** of the crimp sleeve against the outer jacket **48** of the fiber optic cable **46** such that any space between the outer jacket **48** and the optical fiber **50** is eliminated within the cable **46** and the optical fiber **50** gets pinched against the inner surface of the jacket **48** of the fiber optic cable **46**. As such, the optical fiber **50**, as well as the strength layer **52**, can be anchored relative to the connector housing **32** adjacent the proximal end **42** thereof. The location where the optical fiber **52** itself is crimped to the connector housing **32** may be called the fiber anchor location **51** (see FIG. 2).

(86) Anchoring the optical fiber **50** relative to the proximal end **42** of the connector housing **32** can isolate the movable ferrule assembly **26** from the rest of the fiber optic cable **46** that is not pinched or crimped to the connector housing **32**. This is advantageous because, if the optical fiber **50** were not anchored to the connector housing **32**, in certain instances, the optical fiber **50** may slide within the outer jacket **48**, interfering with the predictability and the repeatability of the macrobending that takes place within the fiber take-up region **100** when the ferrule **28** is forced in a proximal direction. For example, if a long fiber optic cable **46** were to be spooled around a spool structure, the fiber **50** might tend to migrate toward the inner diameter side of the cable within the cable and might move a different distance than the outer jacket **48** itself. If the fiber **50** were to slide within the outer jacket **48** toward the ferrule assembly **26**, that would create extra fiber within the connector, interfering with the predictability of the acceptable macrobending that takes place within the fiber take-up region **100**.

(87) In other instances, for example, if a tensile load was applied to the cable in a proximal direction away from the connector, the outer jacket **48** of the cable **46** might stretch inelastically and the optical fiber **50** could slidably move within the jacket, relative to the jacket, causing a pulling force on the ferrule assembly **26**. Thus, by anchoring the optical fiber **50** to the connector housing **32** adjacent the proximal end **42** through the use of the crimp sleeve **44**, the movable ferrule assembly **26** is isolated from the rest of the fiber optic cable **46** that is not crimped to the connector housing **32**. As such, axial load is not transferred in either direction across the anchor location. The anchor restricts/prevents relative movement between the optical fiber and the jacket at the fiber anchor location. In this way, the portion of the fiber within the connector and the portion of the fiber within the main length of the cable are mechanically isolated from one another. The connector of the present disclosure, thus, can operate as designed and utilize the fiber take-up region **100** to provide for a predictable and a repeatable macrobend when the ferrule is moved in a proximal direction relative to the connector housing **32**.

(88) FIGS. **60-65** illustrate two different embodiments of crimp sleeves **544**, **644** that include annular ribs on an exterior surface of the stub portions thereof. Even though the other embodiments of the crimp sleeves disclosed in the present application can be used to crimp the stub portion

thereof against the outer jacket **48** of the fiber optic cable **46** such that the optical fiber **50** gets pinched against the inner surface of the jacket **48** of the fiber optic cable **46**, the crimp sleeves **544** and **644** shown in FIGS. **60-65** may provide for additional material for the stub portions of the crimp sleeve at spots or regions that might tend to deform when the crimp sleeve is crimped at the stub portion.

(89) In the embodiment of the crimp sleeve **544** shown in FIGS. **60-62**, the stub portion **542** of the sleeve **544** includes a first annular rib **543** at a proximal end **547** thereof and a second annular rib **545** at an intermediate location between the proximal end **547** and the radial in-step **541** of the crimp sleeve **544**.

(90) In the embodiment of the crimp sleeve **644** shown in FIGS. **63-65**, the stub portion **642** of the sleeve **644** includes a single, wider annular rib **643** at a proximal end **647** thereof.

(91) In the depicted embodiment, the fiber anchor location is defined as being at a location that is not at a splice location where two segments of optical fiber are spliced together. In the present disclosure, the optical fiber is directly terminated in the connector and the connector is not a splice-on connector.

(92) To assemble the fiber optic connector **20**, the ferrule assembly **26** is first loaded into the distal housing portion **34** of the connector housing **32**. Next, the proximal housing portion **36** is connected to the distal housing **34** (e.g., by a snap fit connection) such that the ferrule hub **30** and the spring **31** are captured within the connector housing **32** at a location between the distal housing portion **34** and the proximal housing portion **46**. Next, an epoxy needle is inserted through the proximal end **42** of the proximal housing portion **36** and is used to inject epoxy into the fiber passage defined through the ferrule **28**. Once the epoxy has been applied, the epoxy needle is removed and the insertion cap **40A** or the insertion cap **40B** is inserted into the proximal end **42** of the connector housing **32**. Thereafter, the strain relief boot **56** and the crimp sleeve **44** are inserted over the fiber optic cable **46** and a distal end portion of the cable is prepared.

(93) As part of the cable preparation process, the jacket **48** is stripped from the distal end portion of the optical fiber. Also, the coating layers **94** are stripped from the distalmost portion of the optical fiber **50** intended to be inserted through the passage defined by the ferrule **28**. Moreover, the strength layer **52** is trimmed to a desired length. Once the fiber optic cable **46** has been prepared, the distal end portion of the optical fiber **50** is inserted through the insertion cap **40A** and into the ferrule **28** which has been potted with epoxy. During the insertion process, the transition **104b** assists in guiding the distalmost end portion of the optical fiber **50** into the ferrule **28**. Once the fiber insertion process has been completed, the crimp sleeve **44** is slid distally over the proximal end **42** of the connector housing **32** and used to crimp the distal end of the strength layer **52** about the exterior surface of the connector housing **32** adjacent to the proximal end **42**. The strain relief boot **56** is then slid distally over the crimp sleeve **44** and proximal end **42** of the housing **32**. Finally, the release sleeve **38** is inserted over the distal end **22** of the fiber optic connector **20** and snapped into place over the connector housing **32**.

(94) Referring to FIGS. **11-13**, the strain relief boot **56** of the fiber optic connector **20** includes a distal end **200** and an opposite proximal end **202**. The strain relief boot defines an inner passage **204** that extends through the boot from the proximal end **202** to the distal end **200**. When the boot **56** is mounted on the connector housing **32**, the inner passage **204** aligns with the central longitudinal axis **70** of the fiber optic connector **20**. The boot **56** includes a connection portion **206** positioned adjacent the distal end **200** and a tapered, strain relief portion **208** positioned adjacent the proximal end **202**. The connection portion **206** has a larger cross-dimension than a corresponding cross-dimension of the tapered, strain relief portion **208**. A transition portion **210** is positioned between the connection portion **206** and the tapered, strain relief portion **208**. An outer surface of the transition portion provides a gradual increase in cross-dimension as the outer surface extends from the tapered, strain relief portion **208** to the connection portion **206**. The outer surface of the transition portion **210** can be pushed to facilitate inserting the connection portion **206** over

the proximal end **42** of the connector housing **32** during assembly of the fiber optic connector **20**. Further details about the boot **56** are provided in U.S. Provisional Patent Application Ser. No. 61/452,935, which is entitled STRAIN RELIEF BOOT FOR A FIBER OPTIC CONNECTOR, and which has been filed on a date concurrent with the filing of the present application.

(95) For the connector **20**, the proximal housing portion **36**, the insertion cap **40A** and the insertion cap **40B** are all depicted as machined metal parts. FIGS. **14-24** show various parts of another fiber optic connector **20'** in accordance with the principles of the present disclosure. The connector **20'** has been modified with respect to the connector **20** so as to include a proximal housing portion **36'**, an insertion cap **40A'** and an insertion cap **40B'** which are all made of molded plastic. The other components of the connector **20'** are the same as the connector **20**. In FIG. **15**, the insertion cap **40B'** is shown installed within the connector **20'**, and a protective outer tube **149** is shown protecting the portion of the coated optical fiber **50** that extends from the proximal side of the ferrule to the boot. The proximal housing portion **36'** is formed by two molded half-pieces **36a** that mate together to form the proximal housing portion **36'**. The half-pieces **36a** can be bonded together with an adhesive or held together mechanically by one or more fasteners such as crimps. According to certain embodiments, the half-pieces **36a** may be held together by a snap-fit interlock. According to the example embodiment depicted in FIGS. **14-24**, each half piece **36a** includes flexible cantilever arms **41** on one side **43** of the half-piece **36a** and notches **45** on the radially opposite side **47** of the half-piece **36a** (see FIGS. **16-17**). Each cantilever arm **41** defines a tab **49** at the end of the arm **41** that is configured to snap over shoulders **51** defined at the notches **45** when two half-pieces **36a** are interlocked together. The cantilever arms **41** and the notches **45** of one half-piece **36a** are provided on opposite sides with respect to the arms **41** and notches **45**, respectively, of the other half-piece **36a**. As such, when the two half-pieces **36a** are brought together for a snap-fit interlock, the cantilever arms **41** of one half-piece **36a** align with the notches **45** of the opposing half-piece **36a** and vice versa.

(96) The molding process used to manufacture the proximal housing portion **36'** allows the interior of the proximal housing portion **36'** to be provided with a continuous curve **150** that extends along the length of the take-up region of connector **20'**. The insertion caps **40A'** and **40B'** are similar to the insertion caps **40A**, **40B** except the parts are molded plastic parts with the inner diameter transitions at the proximal and distal ends of the caps have a more curved profile.

(97) FIGS. **27** and **28** illustrate a prior art fiber optic connector **220** in the form of a conventional LC connector. As shown in FIGS. **27** and **28**, the conventional LC connector **220** includes a connector housing **222** defining a distal housing portion **224** and a proximal housing portion **226**. The LC connector **220** includes a ferrule assembly **228** defined by a ferrule **230**, a hub **232**, and a spring **234**. A proximal end **236** of the ferrule **230** is secured within the ferrule hub **232**. When the LC connector **220** is assembled, the ferrule hub **232** and the spring **234** are captured between the distal housing portion **224** and the proximal housing portion **226** of the connector housing **222** and a distal end **238** of the ferrule **230** projects distally outwardly beyond a distal end **240** of the connector housing **222**. The spring **234** is configured to bias the ferrule **230** in a distal direction relative to the connector housing **222**.

(98) According to certain embodiments, the distal housing portion **224** may be formed from a molded plastic. The distal housing portion **224** defines a latch **242** extending from a top wall **244** of the distal housing portion **224** toward the proximal end **246**, the latch **242** extending at an acute angle with respect to the top wall **244** of the distal housing portion **224**. The distal housing portion **224** also includes a latch trigger **248** that extends from the proximal end **246** of the distal housing portion **224** toward the distal end **240**. The latch trigger **248** also extends at an acute angle with respect to the top wall **244**. The latch trigger **248** is configured to come into contact with the latch **242** for flexibly moving the latch **242** downwardly.

(99) As is known in the art, when the fiber optic connector **220** is placed in an LC adapter **250** for optically coupling light from two optical fibers together, the latch **242** functions to lock the fiber

optic connector **220** in place within the adapter **250**. The fiber optic connector **220** may be removed from the adapter **250** by depressing the latch trigger **248**, which causes the latch **242** to be pressed in a downward direction, freeing catch portions **252** of the latch **242** from the fiber optic adapter **250**.

(100) The region of the distal housing portion **224** from where the latch trigger **248** extends defines a pin hole **254**. The pin hole **254** is configured to receive a pin for forming a duplex LC connector by coupling two simplex connectors **220** in a side-by-side orientation.

(101) Still referring to FIGS. **27** and **28**, a strain relief boot **256** is slid over a proximal end **258** of the proximal housing portion **226** and snaps over a boot flange **260** to retain the boot **256** with respect to the connector housing **222**. The proximal end **258** of the proximal housing portion **226** defines a crimp region **262** for crimping a fiber optic cable's strength layer to the proximal housing portion **226**, normally with the use of a crimp sleeve (not shown). The exterior surface **264** of the proximal housing portion **226** defining the crimp region **262** can be textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the crimp on the housing **222**.

(102) As discussed above with respect to the embodiments of the SC connector shown in FIGS. **1-26**, movement of the ferrule **230** of the LC connector in a proximal direction relative to the connector housing **222** causes the optical fiber to be forced/displaced in a proximal direction relative to the connector housing **222** and the jacket of the fiber optic cable. However, in the conventional LC connector **220** shown in FIGS. **27** and **28**, the passage **266** defined by the proximal housing portion **226** that extends along the longitudinal axis of the connector **220** defines a generally uniform inner diameter DLC similar in size to the diameter of the portion of the optical fiber that includes the core, the cladding layer and the one or more coating layers. As such, the proximal housing portion **226** of a conventional LC connector **220** does not include a fiber take-up region to prevent signal degradation related to microbending caused by the axial displacement of the optical fiber in the proximal direction.

(103) FIGS. **29-45** illustrate various parts of a third fiber optic connector **300** in accordance with the principles of the present disclosure. The connector **300** includes inventive features similar to those shown and described for the SC type connectors **20**, **20'** of FIGS. **1-26**, however, is provided in an LC connector footprint.

(104) Referring to FIGS. **29-45**, the fiber optic connector **300** includes a connector housing **301** including a distal housing portion **302** and a proximal housing portion **304**. The distal housing portion **302** is similar in configuration to that of a conventional LC connector and includes a ferrule assembly **306** defined by a ferrule **308**, a hub **310**, and a spring **312** mounted therein. The ferrule hub **310** and the spring **312** are captured within the distal housing portion **302** by the proximal housing portion **304** of the connector housing **301**. The distal housing portion **302** defines slots **314** that are configured to receive ribs **316** formed at a distal end **318** of the proximal housing portion **304** for snap-fitting the two housing portions **302**, **304** together.

(105) An insertion cap **320** having features similar to insertion caps **40A** and **40A'** is inserted into a proximal end **322** of the proximal housing portion **304**. As discussed above with respect to the SC style connectors **20**, **20'**, an alternative embodiment of an insertion cap having a larger inner diameter for accommodating a protective tubing can also be used. A crimp sleeve **324** is inserted over the proximal end **322** of the proximal housing portion **304** and captures the insertion cap **320** thereagainst. The crimp sleeve **324** is used to crimp a fiber optic cable in a manner similar to that described above for the SC style connectors **20**, **20'**.

(106) A strain relief boot **326** is mounted over the proximal end **322** of the proximal housing portion **304**. The strain relief boot **326** includes a connection portion **328** defining a generally circular inner passage **330** (see FIGS. **44** and **45**). An annular inner lip **332** defined at a distal end **334** of the strain relief boot **326** mounts over a generally round boot flange **336** defined on the outer surface **338** of the proximal housing portion **304**. When the strain relief boot **326** is mounted over the proximal housing portion **304**, the distal end **334** of the strain relief boot **326** abuts against

a stop ring **340**. As shown in FIG. 33, the stop ring **340** defines a conical configuration **342** along the longitudinal direction of the connector **300**, the ring **340** tapering down as it extends from a proximal end **344** toward a distal end **346**.

(107) When the fiber optic connector **300** is fully assembled, the connector **300** retains the overall outer dimension of a conventional LC connector such that two fiber optic connectors **300** can be mounted side by side in a standard duplex configuration. FIGS. 37 and 38 illustrate two of the fiber optic connectors **300** mounted together using a duplex clip **348**. FIGS. 34-36 illustrate two of the fiber optic connectors **300** mounted in a standard duplex LC adapter **250** in a side by side configuration.

(108) As noted above, as shown in FIGS. 33, 42, and 43, the proximal housing portion **304** and the insertion cap **320** of the connector **300** are configured to provide a fiber take-up spacing **350** for allowing macrobending of the optical fiber within the connector housing **301**, in a similar fashion to that described above for the SC style connectors **20**, **20'**. For the connector **300**, the proximal housing portion **304** and the insertion cap **320** are depicted as machined metal parts.

(109) FIGS. 46-59 illustrate various parts of a fourth embodiment of a fiber optic connector **400** in accordance with the principles of the present disclosure. The connector **400** has been modified with respect to the connector **300** so as to include a proximal housing portion **402** and an insertion cap **404** which are made of molded plastic. In addition, unlike the proximal housing portion **304** of the connector **300** described above, which has a fiber take-up region **350** defined by a circular passage **352** extending from the proximal end **322** of the proximal housing portion **304** to the distal end **318** thereof, the proximal housing portion **402** of the connector housing **406** defines an obround passage **408** that transitions to a generally circular passage **410** as it extends from a proximal end **412** of the proximal housing portion **402** to the distal end **414** thereof. As shown in FIG. 54, the passage defines an obround configuration **408** from the proximal end **412** until it reaches the transition portion **416** coming before the neck portion **418**. The obround portion **408** of the passage is provided to increase the predictability of the bending of the fiber as the fiber is exposed to axial displacement within the connector **400** and control the direction of the bend.

(110) As shown in the cross-sectional views provided in FIGS. 52 and 53, the obround portion **408** of the passage defines a larger cross-dimension CDO along a first direction DO1 (taken along lines 55-55 of FIG. 54) than a second direction DO2 (taken along lines 53-53 of FIG. 52). In addition, by providing an obround internal passage **408**, the size of the opening **420** at the proximal end **412** of the proximal housing portion **402** is increased relative to the annular circular opening **354** of the connector **300** shown in FIGS. 29-45 when that opening **420** is measured along the longer cross dimension CDO of the obround passage **408**. By providing an obround passage **408**, the sidewall **422** defined along the longer cross dimension CDO of the obround passage **408** is able to be decreased relative to a uniform sidewall **356** that is provided about the circular opening **354** of the connector **300**.

(111) The insertion cap **404** of the connector **400** defines a stub portion **426** having an exterior obround configuration **428** to match that of the proximal end **412** of the proximal housing portion **402**. As shown in FIGS. 56-59, the insertion cap **404** also defines an internal passage **430** that transitions from a generally circular opening **432** to an obround configuration **434** as the passage **430** extends from the proximal end **436** to the distal end **438** of the insertion cap **404**. The obround portion **434** of the passage **430** cooperates with the obround portion **408** of the internal passage of the proximal housing portion **402** in controlling the direction of the fiber bend.

(112) Although in the foregoing description, terms such as “top”, “bottom”, “front”, “back”, “rear”, “right”, “left”, “upper”, and “lower” may have been used for ease of description and illustration, no restriction is intended by such use of the terms. The connectors described herein can be used in any orientation, depending upon the desired application.

(113) The above specification, examples and data provide a description of the inventive aspects of

the disclosure. Many embodiments of the disclosure can be made without departing from the spirit and scope of the inventive aspects of the disclosure.

Claims

1. A fiber optic connector comprising: a connector housing defining a distal end, a proximal end, and a central longitudinal axis, wherein a ferrule terminating an optical fiber is disposed adjacent the distal end of the connector housing; and a fiber passage extending between the distal end and the proximal end of the connector housing, wherein the fiber passage of the fiber optic connector has a fiber take-up region that is configured to take up an excess length of the optical fiber within the fiber passage due to bending of the optical fiber when the ferrule is moved in a proximal direction away from the distal end, the fiber take-up region defined at least in part by a smooth cavity wall that gradually curves away from the central longitudinal axis of the connector housing as the fiber passage transitions into the fiber take-up region in at least the proximal direction away from the distal end and a distal direction away from the proximal end, wherein at least a portion of the smooth cavity wall gradually curves away from the central longitudinal axis of the connector housing at an increasing slope in at least the proximal direction away from the distal end and the distal direction away from the proximal end.
2. A fiber optic connector according to claim 1, wherein the fiber take-up region is defined by a smooth cavity wall that gradually curves away from the central longitudinal axis so as to form a concave surface in at least one transverse direction as the fiber passage transitions into the fiber take-up region in both the proximal direction away from the distal end and the distal direction away from the proximal end.
3. A fiber optic connector according to claim 2, wherein the fiber take-up region is defined by smooth cavity walls that curve away from the central longitudinal axis in two opposing transverse directions as the fiber passage transitions into the fiber take-up region in both the proximal direction away from the distal end and the distal direction away from the proximal end.
4. The fiber optic connector of claim 1, further comprising a fiber optic cable carrying the optical fiber that is terminated to the ferrule, the fiber optic cable including a cable jacket surrounding the optical fiber, the fiber optic cable also including a strength layer positioned between the optical fiber and the cable jacket, the strength layer being anchored to the connector housing, the optical fiber extending through the fiber passage of the fiber optic connector from the proximal end of the connector housing to the ferrule.
5. The fiber optic connector of claim 4, wherein the cable jacket has an outer diameter less than 1.5 millimeters.
6. The fiber optic connector of claim 5, wherein the outer diameter of the cable jacket is less than or equal to 1.2 millimeters.
7. The fiber optic connector of claim 1, wherein the smooth cavity wall of the fiber take-up region that gradually curves away from the central longitudinal axis of the connector housing as the fiber passage transitions into the fiber take-up region is positioned proximally offset from the distal end of the connector housing.
8. The fiber optic connector of claim 1, wherein the connector housing is an inner housing of the fiber optic connector that further includes a sleeve mounted over the connector housing.
9. The fiber optic connector of claim 1, wherein the fiber optic connector further comprises a strain relief boot coupled adjacent the proximal end of the connector housing.
10. The fiber optic connector of claim 1, wherein the ferrule is movable in a proximal direction relative to the connector housing from a distal position to a proximal position, the distal and proximal positions being separated by an axial displacement distance, wherein the axial displacement distance is at least .015 inches.

11. The fiber optic connector of claim 10, wherein the axial displacement distance is at least .025 inches.
